MDM – ICP Project Report

on

Smart House Fire Detection and Alert System

Flame-Sensor-Based Fire Detection with Automatic Pump-Driven Water Suppression

Submitted in partial fulfilment of the requirements of the degree of

Bachelor of Technology

In

Computer Engineering

by

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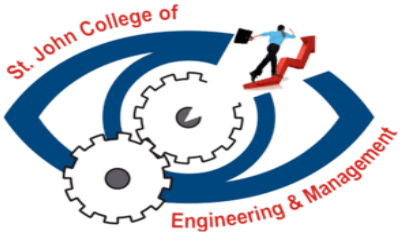
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Abstract

Fire is a serious hazard in homes and other buildings, and early warning can help people react sooner. This synopsis presents the **Smart House Fire Detection and Alert System**, a low-cost educational prototype built on a cardboard smart-house model. An Arduino Uno R3 reads a flame sensor (digital output, active LOW in our setup) and a DHT11 temperature and humidity sensor. When the flame signal is triggered, the system switches on an LED and a buzzer, shows a fire alert on a 16×2 I2C LCD, and uses a servo motor to move a low-voltage rocker switch that connects a small submersible water pump to its own separate battery supply, demonstrating water delivery through a tube. When the signal clears, the servo returns the switch to the OFF position. Temperature and humidity are displayed for monitoring only and are not used as a fire trigger. The program is written in C++ in the Arduino IDE using the Servo, DHT and LiquidCrystal_I2C libraries.

The report covers the background, a review of ten verified research papers, the proposed methodology, project planning, feasibility, expected results, sustainability and conclusion. During development a relay was replaced by the servo-operated switch after intermittent operation, and LCD display corruption was observed when the pump ran near the electronics; the display worked correctly when the pump was kept away (the root cause was not confirmed). The project is an educational integration of existing components. It is not a new invention and is not certified for real-world fire protection.

Keywords: Arduino Uno, flame sensor, DHT11, fire alert, servo-operated switch, water pump, embedded systems, educational prototype.

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Abbreviations and Symbols

Abbreviation / Symbol	Meaning
BIS	Bureau of Indian Standards
BOM	Bill of Materials
DC	Direct Current
DHT11	Digital-output temperature and humidity sensor (model name)
EMI	Electromagnetic Interference
GPIO	General-Purpose Input/Output
GSM	Global System for Mobile Communications
I2C	Inter-Integrated Circuit (two-wire serial bus)
IDE	Integrated Development Environment
IoT	Internet of Things
IR	Infrared
LCD	Liquid Crystal Display
LED	Light-Emitting Diode
PWM	Pulse-Width Modulation
RH	Relative Humidity
SCL / SDA	Serial Clock / Serial Data lines of the I2C bus
SDG	Sustainable Development Goal
SMS	Short Message Service
USB	Universal Serial Bus
WBS	Work Breakdown Structure
°C	degree Celsius (symbol)
°	degree of servo shaft angle (symbol)
V	volt (symbol)
%	per cent (symbol)

CHAPTER 1: INTRODUCTION

1.1 Background

Fire can spread quickly inside a building, so the time between the start of a fire and the moment people become aware of it is very important. An automatic fire detection and alarm system is meant to detect a fire at the earliest practicable moment and give a warning; it can also be arranged to start other actions, such as the operation of a fire-extinguishing system [1]. Commercial and industrial systems are designed, installed and maintained according to standards such as the Indian Standard IS 2189 [1].

For students, the same ideas can be studied on a small scale with a **microcontroller** (a small programmable computer on a single chip), **sensors** (devices that convert a physical quantity into an electrical signal) and **actuators** (devices such as buzzers, motors and pumps that act on the outside world). The Arduino Uno R3 is a popular open-source board built around the ATmega328P microcontroller; it offers 14 digital input/output pins, 6 analog inputs and I2C, SPI and UART communication, and is widely used for education and prototyping [2]. Many researchers have therefore built Arduino-based or IoT-based fire alarms, with features ranging from a simple buzzer to SMS alerts, mobile apps and automatic sprinklers [3]– [12].

This project builds a physical **model** in which an Arduino Uno detects a flame-related signal, gives a visual and audible warning, displays temperature and humidity on a 16×2 I2C LCD, and switches a small water pump using a servo-operated rocker switch. The model is an educational prototype that shows how sensing, alerting, display and a basic automatic response can be combined in one low-cost system.

1.2 Problem Statement

The problem addressed by this project is stated below under the headings suggested in the AF 52 format.

- **Existing problem:** Fire detection involves several parts (sensing, alerting, display and a response mechanism). Students rarely get a compact, low-cost physical model on which all of these can be connected, programmed and tested together.
- **Industrial issue:** Certified fire detection and alarm systems must follow standards and approved equipment [1]; they are not meant for classroom experiments, and a student prototype must never be treated as a replacement for them.
- **Research gap:** Several reviewed papers concentrate on alarm and notification, or on suppression through sprinklers or robots [3]– [12]. In the reviewed abstracts, little attention is given to a very simple standalone build that also documents practical integration problems such as pump switching and electrical interference (see Chapter 2).

- **Need improvement:** A simple, reliable way of switching a separately powered pump from a small microcontroller, together with clear local alerts that do not depend on the Internet.

Problem statement: To design, build and test a low-cost Arduino Uno–based smart-house model that detects a flame-related signal, gives local visual and audible alerts, displays temperature and humidity, and demonstrates a basic automatic water-pump response through a servo-operated rocker switch, with the pump powered from a separate supply.

1.3 Need of the Project

- **Low cost and availability:** All components are inexpensive hobby-level parts that are easy to obtain and replace.
- **Learning by integration:** Connecting sensors, an LCD, indicators, a servo and a pump teaches wiring, programming and testing in one project.
- **Local alerts:** The LED, buzzer and LCD work without any network, which keeps the design simple and easy to demonstrate.
- **Basic automatic response:** The servo-operated switch shows how a small controller can start a separately powered device without driving it directly from a GPIO pin.
- **Recorded lessons:** Problems met during development (relay behaviour, servo direction, display corruption near the pump) are documented so that other students can avoid them.

1.4 Objectives

General objective

To develop a low-cost, educational smart-house fire detection and alert prototype using an Arduino Uno that demonstrates detection, local warning, environmental monitoring and a basic automatic water-pump response.

Specific objectives

1. Design the system around an Arduino Uno R3, a flame sensor, a DHT11 sensor, a 16×2 I2C LCD, an LED, a buzzer module and a servo motor.
2. Develop a C++ program in the Arduino IDE using the Servo, DHT and LiquidCrystal_I2C libraries to read the sensors and control the outputs.
3. Fabricate a cardboard smart-house model and mount the components on it, with the pump and its separate battery kept apart from the controller electronics.
4. Test each component individually, then integrate and test the complete system.

5. Demonstrate the servo-operated rocker switch that connects the separately powered pump and delivers water through a tube.
6. Record the problems observed during development and the changes made to solve them.

1.5 Scope

What will be covered

- Detection of a flame-related digital signal from a flame sensor (active LOW in our setup).
- Local warning by LED, buzzer and a fire alert message on the LCD.
- Display of temperature and relative humidity from the DHT11 sensor for environmental monitoring.
- Operation of a low-voltage rocker switch by a servo motor (ON = 60°, OFF = 120°) to switch a small pump powered from a separate battery holder.

Limitations

- The system depends on a single flame sensor signal. It has no smoke or gas sensor, so it can miss a fire that gives no flame signal and can react to other sources that the sensor responds to.
- The DHT11 readings are displayed only; the current program does not use temperature as a separate fire-detection trigger.
- The pump shows water delivery through a tube in a model. The prototype is **not claimed to extinguish real fires**, and it is **not certified** for real-world fire protection.
- No network alert (SMS, mobile app or web page) is included, and no events are logged.

Future expansion

Possible improvements, clearly marked as future work and not as current features, are listed in Section 7.5.

1.6 Expected Outcomes

- **Working prototype:** a cardboard smart-house model with the sensing, alert, display and pump-switching hardware integrated.
- **Software:** an Arduino sketch (C++) that implements the working principle described in Chapter 3.
- **Documentation:** this synopsis, the connection diagram and the development notes.
- **Simulation and CAD model:** not applicable; the project is built directly as a physical model.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Research on low-cost fire detection falls into a few groups: (a) Arduino or microcontroller alarms with a local buzzer or display; (b) IoT and GSM systems that send SMS messages, calls or mobile-app notifications; (c) multi-sensor systems that combine flame, smoke, gas, heat and humidity sensing; and (d) systems that add an automatic sprinkler, pump or fire-fighting robot. This chapter summarises ten research papers from these groups.

Each paper listed in Table 1 was checked against its publisher or repository record (title, authors, year, venue and, where available, DOI). The summaries are based on the published abstracts and records, not on a full reading of every paper, and the "Research Gap" column describes what the abstract does not mention in relation to our project. It is not a criticism of the work. The references are listed in IEEE style at the end of this report.

2.2 Literature Review Table

Table: Literature review of Arduino-based and IoT fire detection systems

Paper	Author	Year	Method	Findings	Research Gap
IoT-based fire alarm system [3]	A. Mahgoub, N. Tarrad, R. Elsherif, A. Al-Ali, L. Ismail	2019	Wireless nodes (ESP8266 NodeMCU) with smoke, temperature, humidity, flame, methane and CO sensors; central Raspberry Pi with a 4G module; MQTT messaging.	Sends SMS to the fire department and user, calls the user and sounds a local alarm. A prototype worked with an average delay of under 30 s.	Network of many sensors plus 4G module is complex and costly for a basic classroom model; no water-based response.
Design of a smart independent smoke sense system based on NB-IoT technology [4]	J. Wang, J. Su, R. Hua	2019	Stand-alone smoke, temperature and humidity alarm using NB-IoT and an STM32 chip with the CoAP protocol.	Low cost and very low power; reported standby of about 5 years after testing; multi-party alarm.	Needs a specialised IoT network and chip; alarm only, no automatic response.
Arduino based fire detection and control system [5]	M. S. Dauda, U. S. Toro	2020	Arduino-based automatic fire alarm detection and control system, organised into several modules.	Proposes the Arduino as the base of a stand-alone, low-cost fire detection and control solution for homes, offices and industries.	Abstract-level review only; it does not describe isolating the pump circuit or a servo-operated switch.
A smart fire detection system using IoT technology with automatic water sprinkler [6]	H. Alqourabah, A. Muneer, S. M. Fati	2021	Flame, heat and smoke sensors; GSM modem; Ubidots IoT platform; automatic water sprinkler; aims to reduce false alarms.	Authors report good affordability, effectiveness and responsiveness.	Uses GSM, cloud platform and several sensors; more complex than a simple educational build.
A multisensory Arduino-based fire detection and alarm system using GSM and RF module with an Android application [7]	M. C. Baba, J. B. Grado, D. J. L. Solis, I. M. Roma, J. T. Dellosa	2022	Arduino Uno with flame, smoke, butane and temperature sensors; SMS and an Android app; three detection levels.	Alerts are sent to the user's phone; the authors report that the device performed well when tested.	Depends on communication modules; no water-based response.

Paper	Author	Year	Method	Findings	Research Gap
Smart fire detection system with call alert and water sprinkler unit using IoT [8]	R. Ranjith, S. L. Naveen	2022	IoT fire detection with a smart water sprinkler for the fire-cracker industry; call and SMS alerts with GPS co-ordinates.	Triggers the water system while sending alerts, aiming to avoid hazardous fire accidents.	Industry-oriented and needs cellular and GPS services; not a simple stand-alone student model.
Fire detection and water discharge activity for fire fighting robots using IoT [9]	M. A. Al Rakib et al.	2022	Arduino Uno robot in which a flame sensor activates a water pump and a servo motor.	Prototype detects fire and sprays water at about 20 cm from the fire.	Robot-focused; no temperature or humidity display; pump power isolation is not highlighted in the abstract.
A sustainable and efficient IoT approach for fire department alerting system [10]	C. Vidyadhari, P. Pavan, M. Tharun, V. Kundu, K. Dulta	2023	Arduino board, flame sensor, buzzer and SIM800L module; the flame sensor responds to infrared radiation.	Gives a local buzzer alarm and sends SMS or calls to preset contacts; easy to extend with more sensors.	Alert only; no automatic water response or environmental display.
Design and construction of an Arduino-based smoke detection system [11]	A. O. Ilori, S. Emmanuel, O. Y. Omogunloye	2025	Arduino Uno with an MQ2 smoke/gas sensor, LCD, LEDs and buzzer; tested in low, moderate and high smoke cases.	Authors report consistent detection and rapid response in the three test cases.	Smoke sensing only; no flame input and no suppression mechanism.
IoT-based fire automatic surveillance system with sprinkler control systems [12]	M. F. Lehon, R. Mohd Zin	2026	ESP8266 NodeMCU with IR flame, MQ-2 and BMP280 sensors; buzzer and water spray; Blynk app notifications.	Reported average response time of 7.13 s and cumulative detection accuracy of 66.91 %, which suggests that low-cost sensing has limits.	Depends on Wi-Fi and an IoT platform; its reported accuracy shows why we avoid strong accuracy claims.

2.3 Research Gap

What previous researchers have done

The reviewed papers show that fire detection with microcontrollers is well established. Researchers have used flame, smoke, gas, heat and humidity sensors [3], [4], [6], [7], [11]; added GSM, Wi-Fi, NB-IoT or mobile-app alerts [3], [7], [8], [10], [12]; and added water sprinklers, pumps or robots for a response [6], [8], [9], [12].

What remains less covered

Many of these systems rely on communication modules, cloud platforms or several sensors, which makes them harder to rebuild in a basic laboratory. The abstracts also say little about practical points that students meet when they add a pump to a small controller, such as how to switch a separately powered pump reliably and how pump operation can disturb nearby electronics.

How this project addresses the gap

Our project is a simple stand-alone build with local alerts only. It keeps the pump on its own supply and uses a servo to operate a rocker switch, after a relay gave intermittent operation. It also records the problems seen during development. The aim is a clear, reproducible educational example, not better detection performance.

2.4 Novelty / Innovation

Fire detection already exists, and this project is not a new invention. Its value is an educational integration of available components into one working, low-cost model.

Unique features of this implementation

- A servo motor physically moves a low-voltage rocker switch, so the pump circuit stays electrically separate from the Arduino and its GPIO pins.
- Local alerts (LED, buzzer, LCD) together with live temperature and humidity display, with no network needed.
- A documented development history, including the change from a relay to a servo-operated switch and the interference observed near the pump.

Improvements over the reviewed systems, and what we do not claim

Compared with the network-based systems above [3], [6], [7], [8], [12], our design is simpler, cheaper to rebuild and easier to explain. We do **not** claim better accuracy, faster response or suitability for real fire protection; the prototype has not been compared with the reviewed systems on any measured performance.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System

The proposed system is a small Arduino Uno–based fire detection and alert prototype mounted on a cardboard smart-house model. It has three parts:

- **Inputs:** a flame sensor whose digital output goes LOW when the flame-related signal is triggered, and a DHT11 sensor that supplies temperature and relative humidity.
- **Processing and outputs:** the Arduino Uno R3 reads the inputs and controls an LED, a buzzer module and a 16×2 I2C LCD that shows the status and the environmental readings.
- **Response mechanism:** a servo motor physically moves a low-voltage rocker switch. The switch connects a small DC submersible pump to its own separate battery holder, so water can be delivered through a tube as a demonstration.

The pump is **not** powered from the Arduino or from any GPIO pin; only the servo, which is a signal-controlled part of the Arduino circuit, moves the switch mechanically. A relay was tried first but showed intermittent operation, so the final design uses the servo-operated switch.

3.2 Working Principle

1. The Arduino is powered through its USB cable. On start-up the program initialises the LCD, the DHT11 sensor, the input and output pins and the servo, and puts the pump switch in the OFF position (servo angle 120°).
2. The Arduino repeatedly reads the digital output of the flame sensor. In our setup the output is active LOW, so a LOW level means the flame-related signal has been triggered.
3. The Arduino also reads temperature and humidity from the DHT11 sensor.
4. If the flame signal is **detected**, the Arduino switches on the LED and the buzzer and shows a fire alert on the LCD.
5. At the same time the servo moves the rocker switch to the ON position (servo angle 60°). This completes the separate low-voltage pump circuit, and the pump delivers water through the tube.
6. If the flame signal is **not detected** (or after it clears), the LED and buzzer are kept off and the servo moves the switch back to the OFF position (120°), so the pump stops.
7. The LCD is updated with the current status together with temperature and humidity, and the cycle repeats continuously.

The ON and OFF angles are 60° and 120° because the rocker switch orientation was reversed during assembly. Temperature is displayed for monitoring only; it is not used as a separate fire trigger.

3.3 System Architecture / Block Diagram

Figure 1 shows the block diagram. The blue region is the Arduino-controlled circuit; the orange region is the separate pump power circuit. The dashed orange arrow shows the **mechanical** link from the servo to the rocker switch; there is no electrical connection between the Arduino circuit and the pump circuit.

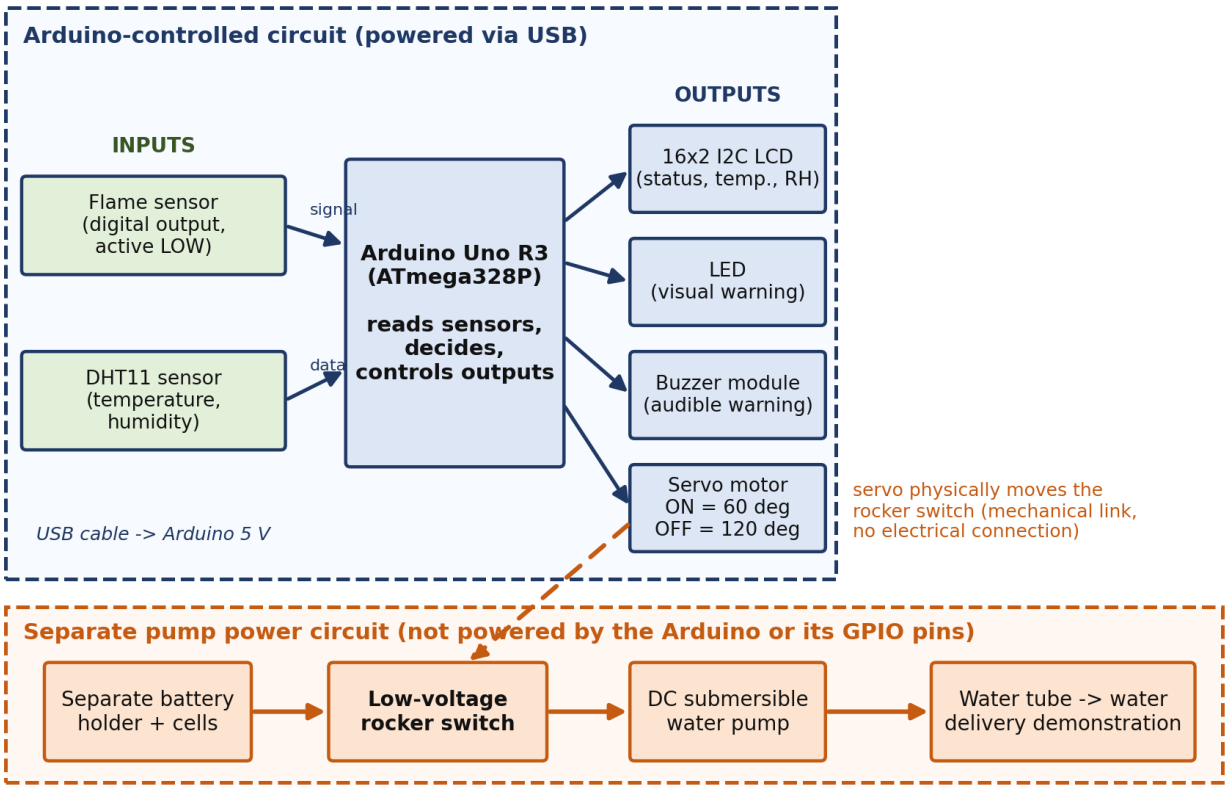


Figure: Block diagram of the smart house fire detection and alert system

Table: Blocks of the system and their functions

Block	Function	Connected to
Arduino Uno R3	Main controller; reads sensors and controls the outputs.	Flame sensor, DHT11, LCD, LED, buzzer, servo (USB powered).
Flame sensor	Gives a digital signal; LOW means the flame-related signal is triggered.	Arduino digital input.
DHT11 sensor	Measures temperature and relative humidity.	Arduino digital input (data line).
16×2 I2C LCD	Shows fire alert, temperature and humidity.	Arduino I2C pins (SDA, SCL).
LED	Visual warning.	Arduino digital output.
Buzzer module	Audible warning.	Arduino output.
Servo motor	Moves the rocker switch: ON = 60°, OFF = 120°.	Arduino signal pin; mechanically linked to the switch.
Rocker switch	Makes or breaks the pump's separate low-voltage circuit.	Battery holder and pump (no link to Arduino pins).

Block	Function	Connected to
Battery holder + pump + tube	Separate supply for the DC submersible pump; tube shows water delivery.	Rocker switch only.

3.4 Flow Chart

Figure 2 shows the program flow: Start → initialize components → read sensors → check the flame signal → if detected, activate the alert and move the servo to pump ON; otherwise keep the alerts off and the pump switch OFF → update the LCD → repeat.

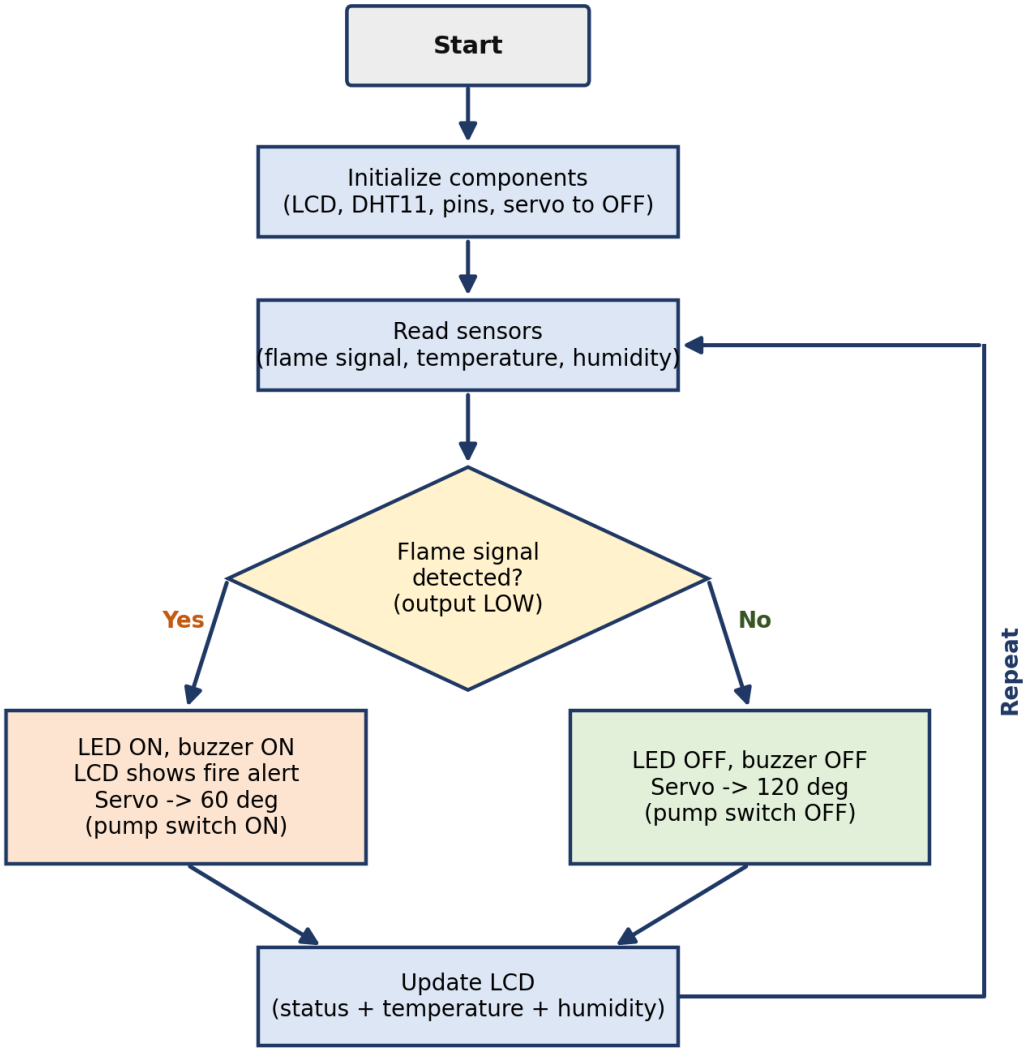


Figure: Flow chart of the program

3.5 Methodology

3.5.1 Design procedure

1. Define the required functions: flame-signal detection, local alerts, temperature and humidity display, and a pump response.
2. Select simple, low-cost components that match these functions (Section 3.6).
3. Decide that the pump will have its own battery supply and be switched by a rocker switch moved by the servo.
4. Plan the layout of the cardboard house so that the pump and its battery are kept apart from the LCD and controller electronics.

3.5.2 Experimental (testing) procedure

Each of the flame sensor, DHT11, LCD, LED, buzzer and servo was first tested individually with a small test program. After each part worked, the parts were combined step by step on the breadboard and then in the cardboard model. Test observations are written as qualitative notes; no performance figures are claimed in this synopsis.

3.5.3 Programming steps

The system is programmed in C++ using the Arduino IDE. The program uses the Servo library to control the servo motor, the DHT library to read the DHT11 sensor and the LiquidCrystal_I2C library to operate the LCD. The program follows these steps:

1. Include the required libraries and define the input and output pins.
2. Initialise the LCD, DHT11 sensor, serial communication and servo motor.
3. Configure the flame sensor as a digital input and the LED and buzzer as outputs.
4. Move the servo to the calibrated OFF position of 120° during initialisation.
5. Continuously read the flame sensor. In the project setup, a LOW digital signal indicates that the flame sensor has been triggered.
6. When the flame signal is detected, turn on the LED and buzzer, display a fire alert on the LCD and move the servo to 60° to operate the rocker switch.
7. When the flame signal clears, turn off the LED and buzzer and return the servo to 120° to switch off the pump.
8. Display the current system status, temperature and humidity on the LCD.

The DHT11 readings are used for environmental monitoring only; they do not independently trigger the fire response. The pump is powered by a separate low-voltage battery supply and is not driven directly by an Arduino pin. The complete program is provided in Appendix C.

3.5.4 Fabrication of the model

The smart-house structure is made of cardboard. The Arduino, breadboard, sensors, LCD, LED and buzzer are mounted on the house, with jumper wires for connections. The servo is positioned so that its arm can push the rocker switch between its two positions. The pump, its separate battery holder and the rocker switch form a separate circuit, and a tube carries the water. Care is taken to keep water away from the electronics.

3.5.5 Development and testing history

The following notes summarise what was done and observed during development. They are development observations, not experimental results.

Table : Development and troubleshooting summary

Stage	Activity	Observation / action taken
1	Individual testing	The flame sensor, DHT11, LCD, LED, buzzer and servo were tested one by one before integration.
2	LCD set-up	An initial LCD display issue was resolved and the I2C LCD was configured.
3	Buzzer	The buzzer was tested and its control in the program was adjusted to give a clearer warning tone.
4	Pump control	A relay was tried first but gave intermittent operation. The design was changed to a servo-operated rocker switch.
5	Servo direction	The ON/OFF direction was wrong because the switch orientation was reversed; the final angles were set to OFF = 120° and ON = 60°.
6	Display corruption	The LCD showed corrupted characters when the pump operated near the electronics. The display worked correctly when the pump was moved away. This is recorded as an observed interference issue; its root cause was not confirmed.
7	Integration	Sensors, display, LED, buzzer, servo and the pump mechanism were integrated into the cardboard smart-house model.

3.6 Components / Materials

Table 4 lists the components used. Entries marked as to be confirmed should be completed from the actual parts.

Table : Components and materials

Component	Specification	Quantity
Arduino Uno R3	ATmega328P microcontroller board, 5 V logic, 14 digital I/O pins, 6 analog inputs, USB powered [2].	1
Flame sensor module	Digital output, active LOW in our setup; such modules commonly respond to infrared radiation from flames [10].	1

Component	Specification	Quantity
DHT11 sensor	Digital temperature and humidity sensor; supplier data list about ±2 °C and ±5 %RH accuracy [13] (check your module's data).	1
16×2 I2C LCD	Character display (2 lines × 16 characters) with I2C interface (SDA, SCL).	1
LED	Visual warning indicator; colour red.	1
Buzzer module	Audible warning; type (active/passive).	1
Servo motor	Operates the rocker switch; ON = 60°, OFF = 120°.	1
Low-voltage rocker switch	Switches the pump circuit; moved mechanically by the servo.	1
DC submersible water pump	Small pump powered from a separate battery supply; rated voltage 3V-6V.	1
Battery holder (with cells)	Separate supply for the pump; 2 x 3.7V cells.	1
Water tube and container	Carries water from the pump for the demonstration.	1 set
Breadboard, jumper wires, USB cable	Connections and power/programming for the Arduino.	1 set
Cardboard house structure	Smart-house model that holds the components.	1

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure

The project is divided into the work packages in Table 5. The numbering is used again in the timeline and the roles table.

Table : Work breakdown structure (WBS)

WBS No.	Work package	Main tasks
1.0	**Research and planning**	1.1 Study fire detection basics and Arduino-based systems; 1.2 review research papers; 1.3 define functions and scope.
2.0	**Component selection and procurement**	2.1 Choose sensors, display, indicators, servo and pump; 2.2 estimate cost; 2.3 purchase components.
3.0	**Model and hardware assembly**	3.1 Build the cardboard house; 3.2 mount components; 3.3 wire the circuit; 3.4 prepare the separate pump circuit.
4.0	**Individual component testing**	4.1 Test flame sensor and DHT11; 4.2 test LCD, LED and buzzer; 4.3 test servo and pump.
5.0	**Software development**	5.1 Write the Arduino program; 5.2 configure the I2C LCD; 5.3 set servo angles and alert logic.
6.0	**Integration and troubleshooting**	6.1 Combine hardware and software; 6.2 correct servo direction; 6.3 resolve pump and display problems.
7.0	**Documentation and presentation**	7.1 Prepare the synopsis; 7.2 prepare diagrams and photographs; 7.3 prepare the demonstration.

4.2 Gantt Chart

Table 6 shows an indicative 9-week semester timeline. The shaded cells show when each work package is active. The weeks are planning values and should be changed to match the team's actual dates.

Table: Indicative semester timeline (Gantt chart)

Work package	W1	W2	W3	W4	W5	W6	W7	W8	W9
1.0 Research and planning	■	■							
2.0 Component selection / procurement		■	■						
3.0 Model and hardware assembly			■	■	■				
4.0 Individual component testing				■	■				
5.0 Software development					■	■			
6.0 Integration and troubleshooting						■	■		
7.0 Documentation and presentation								■	■

4.3 Roles and Responsibilities

Team member names and responsibilities have not been supplied, so placeholders are used. Assign the work packages of Table 5 to the members.

Table: Roles and responsibilities

Student	Responsibility
Kamesh Kirange [EU1242073]	Research, project planning and documentation
Prasanna Lohar [EU1242055]	Arduino programming, sensor integration, troubleshooting
Manthan Meher [EU1242047]	Hardware assembly, component testing, integration and final demonstration
Mrs. Sathiyabama Hanilkumar	Guidance, review of progress and approval of the synopsis.

4.4 Risk Assessment

The risks in Table 8 are rated qualitatively (High, Medium, Low). Some of them were met during development; these are marked "observed".

Table: Risk assessment

Risk	Impact	Mitigation
False flame detection (the sensor may respond to other light or heat sources)	**Medium**	Test in controlled lighting; adjust the sensor sensitivity; state clearly that this is a demonstration, not a certified detector.
Missed detection (only a single flame signal is used; no smoke or gas sensing)	**High**	State this limitation; list smoke or gas sensing as future work.
Electrical interference when the pump runs near the electronics (observed: LCD corruption)	**Medium**	Keep the pump and its wiring away from the LCD and Arduino; keep the pump on a separate supply; the root cause is not confirmed and can be investigated later.
Pump malfunction or intermittent switching (observed with the relay)	**Medium**	Use the servo-operated rocker switch; check the pump supply and tube before each demonstration.
Servo moves in the wrong direction or mis-aligns with the switch (observed)	**Low**	Calibrate ON = 60° and OFF = 120° after mounting; recheck after any change.
Water near electronics (short circuit or damage)	**High**	Place the water container and tube away from the controller; use low-voltage supplies only; keep a cloth ready; supervise demonstrations.
Power-supply problems (weak pump batteries, USB supply interruption)	**Medium**	Use fresh batteries; keep the pump supply separate from the Arduino; check connections.
DHT11 reading errors or slow update (limited sensor)	**Low**	Read at suitable intervals and show the latest valid reading; use the readings for monitoring only.
Over-claiming the system's capability	**High**	Describe the system only as an educational prototype; do not state accuracy or response-time figures that were not measured.
Delay in component delivery or team availability	**Low**	Order early; share tasks according to Table 7.

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The project is technically feasible. The Arduino Uno R3 provides the digital pins and the I2C interface needed for the sensors, display, indicators, and servo [2]. The Arduino IDE and the Servo, DHT and I2C LCD libraries are freely available [14], [15]. Each component was tested individually before integration, and the problems found during integration were solved in the model (Section 3.5.5). The remaining technical limitations are listed in Section 1.5.

5.2 Economic Feasibility

The project is designed to be economical by using commonly available hobby-level electronic components and freely available development software. The Arduino IDE and the libraries required for sensor interfacing, LCD control and servo operation are available without a software licence fee. The estimated hardware cost is presented in Table 9. Actual expenditure may vary depending on the supplier, component quality, shipping charges, and the availability of components from the college laboratory. Components reused from previous projects or obtained from laboratory stock should be recorded separately from newly purchased items.

5.3 Operational Feasibility

The prototype is easy to operate: it is powered through the USB cable, the pump batteries are inserted in the separate holder, and the system runs automatically. Maintenance consists of replacing the pump batteries, refilling the water container and checking the servo–switch alignment. All circuits are low-voltage. The prototype is used for demonstration under supervision and is not meant to be left unattended as a real fire-protection device.

5.4 Budget Estimation

Items refer to the numbers in the Bill of Materials (Table 10). To avoid counting any component twice, costs are written only in this budget table; the BOM lists the technical details.

Table: Budget estimation (to be completed with actual prices)

BOM No.	Item	Quantity	Unit cost (₹)	Cost (₹)
B1	Arduino Uno R3	1	350	350
B2	Flame sensor module	1	35	35
B3	DHT11 sensor	1	60	60
B4	16×2 I2C LCD	1	150	150
B5	LED and resistor (set)	1	10	10

BOM No.	Item	Quantity	Unit cost (₹)	Cost (₹)
B6	Buzzer module	1	35	35
B7	Servo motor	1	120	120
B8	Low-voltage rocker switch	1	20	20
B9	DC submersible water pump	1	100	100
B10	Battery holder with cells (for the pump)	1	200	200
B11	Water tube and container	1 set	20	20
B12	Breadboard, jumper wires	1 set	100	100
	Total (sum of the cost column)			**₹ 1200**

5.5 Bill of Materials (BOM)

The BOM lists each item once, with its specification. Items marked “to be confirmed” must be filled in from the actual parts.

Table: Bill of materials

BOM No.	Component	Specification / remark	Qty
B1	Arduino Uno R3	Main microcontroller board (ATmega328P).	1
B2	Flame sensor module	Digital output, active LOW in our setup.	1
B3	DHT11 sensor	Temperature and humidity; used for display only.	1
B4	16×2 I2C LCD	Shows status, temperature, and humidity.	1
B5	LED and resistor	Visual warning; colour red.	1
B6	Buzzer module	Audible warning; type active / passive.	1
B7	Servo motor	Moves the rocker switch; model SG90 micro servo moto.	1
B8	Low-voltage rocker switch	Switches the pump circuit.	1
B9	DC submersible water pump	Small pump; rated voltage 3V-6V.	1
B10	Battery holder with cells	Separate pump supply; 2 x 3.7V cells.	1
B11	Water tube and container	Water delivery demonstration.	1 set
B12	Breadboard, jumper wires, USB cable	Connections, power and programming.	1 set

CHAPTER 6: EXPECTED RESULTS

This chapter describes what the prototype is **expected** to do. In line with the AF 52 format, no actual experimental results appear in this synopsis, and no accuracy percentages, response times or other measured figures are stated. Measured results, if any, belong in the final project report. The problems noted in Section 3.5.5 are development observations and are not treated as results.

6.1 Expected Prototype and Functions

Table: Expected functions and outcomes of the prototype

Function	Expected outcome	How it can be checked later
Flame-related signal detection	The Arduino recognises the flame sensor's active-LOW output as a detection signal and reacts to it.	Observe the response when the sensor's flame-related input is triggered and when it clears.
Visual and audible warning	The LED lights and the buzzer sounds while the detection signal is present, and both switch off when it clears.	Observe the LED and buzzer during triggering and clearing.
LCD temperature and humidity display	The 16×2 I2C LCD shows the DHT11 temperature and humidity, and shows a fire alert when the signal is detected.	Compare the displayed readings with a reference thermometer/hygrometer, if available.
Servo-operated pump switch	The servo moves the rocker switch to ON (60°) on detection and back to OFF (120°) when the signal clears.	Observe the switch position and pump state in each case.
Water delivery demonstration	The separately powered pump delivers water through the tube while the switch is ON and stops when it is OFF.	Observe water flow from the tube end under supervision.
Integration of hardware and software	Sensors, displays, indicators, servo and pump mechanism work together in the cardboard smart-house model under the Arduino program.	Run the complete sequence several times and note any faults.

6.2 Accuracy, Efficiency and Performance

The AF 52 format lists accuracy, efficiency and performance as items to discuss. For this prototype, these are **not quantified** in the synopsis. The DHT11 is a low-cost sensor with limited accuracy [13], and a single flame-sensor signal cannot represent every kind of fire, so the project makes no claim of reliability beyond a classroom demonstration.

6.3 Simulation and Experimental Results

No simulation is planned, and experimental results are deliberately left out of this synopsis. Any measurements taken later should be recorded in a separate results section of the final report.

CHAPTER 7: SUSTAINABILITY & IMPACT

7.1 Industrial Relevance

Real fire detection and alarm systems combine detectors, alarm devices and, in some installations, signals that start other services such as fire-extinguishing systems [1]. The prototype mirrors this idea (detect, warn, trigger a response) on a very small scale, which helps students understand embedded control and interfacing, skills that are used in industrial automation and building systems. It is not a substitute for approved equipment, and any real installation would have to follow the applicable standard [1].

7.2 Societal Impact

Early warnings can give people more time to react to fire. The prototype itself is only a classroom model, but it can raise awareness of fire safety and show how low-cost electronics can support safer homes. Its main social benefit is educational: it gives students a hands-on example of using technology for safety, and the lessons from its problems can be shared with other students.

7.3 Environmental Impact

- **Positive:** the model uses cardboard and low-power electronics, and the components can be reused in later projects.
- **Resource use:** the water demonstration uses only a small quantity of water, which can be reused from the container.
- **Waste:** used batteries and damaged electronic parts should be disposed of at suitable e-waste or battery collection points.
- **Limits:** the prototype does not reduce environmental harm from real fires on its own; the benefit is indirect, through learning and awareness.

7.4 SDG Mapping

Only goals with a reasonable link to this prototype are ticked. The goals follow the United Nations 2030 Agenda [16].

Table: Sustainable Development Goal (SDG) mapping

Selected	SDG	Connection to this project	Strength
☑	**SDG 9: Industry, Innovation and Infrastructure**	Builds skills in embedded systems and shows low-cost innovation in safety-related technology; relevant to building-safety infrastructure in general.	**Direct**
☑	**SDG 11: Sustainable Cities and Communities**	Fire safety in buildings supports safer cities and communities; the prototype is an educational step toward such systems.	**Direct**
☑	**SDG 4: Quality Education**	Hands-on engineering learning through building, testing and documenting a working prototype.	**Supporting**
☐	SDG 3, 7, 12, 13	Considered but not claimed, because the prototype has no measured effect on health, energy, consumption or climate.	Not claimed

7.5 Future Scope

The following are **future improvements only**. None of them is a feature of the current prototype.

- Add a smoke sensor and a gas sensor to reduce the chance of missing a fire that gives no flame signal.
- Add Wi-Fi or mobile notifications so that alerts reach people who are away from the house.
- Add event logging (time and type of each alert) and a web dashboard to view readings and history.
- Use temperature trends from the DHT11 or a better sensor as an additional check to reduce false detection.
- Replace the demonstration mechanism with a more reliable, professionally designed suppression mechanism and a properly isolated pump driver.
- Study the cause of the display corruption seen near the pump and improve wiring, grounding and layout.

CHAPTER 8: DIAGRAM & STRUCTURE

8.1 Connection (Wiring) Diagram.

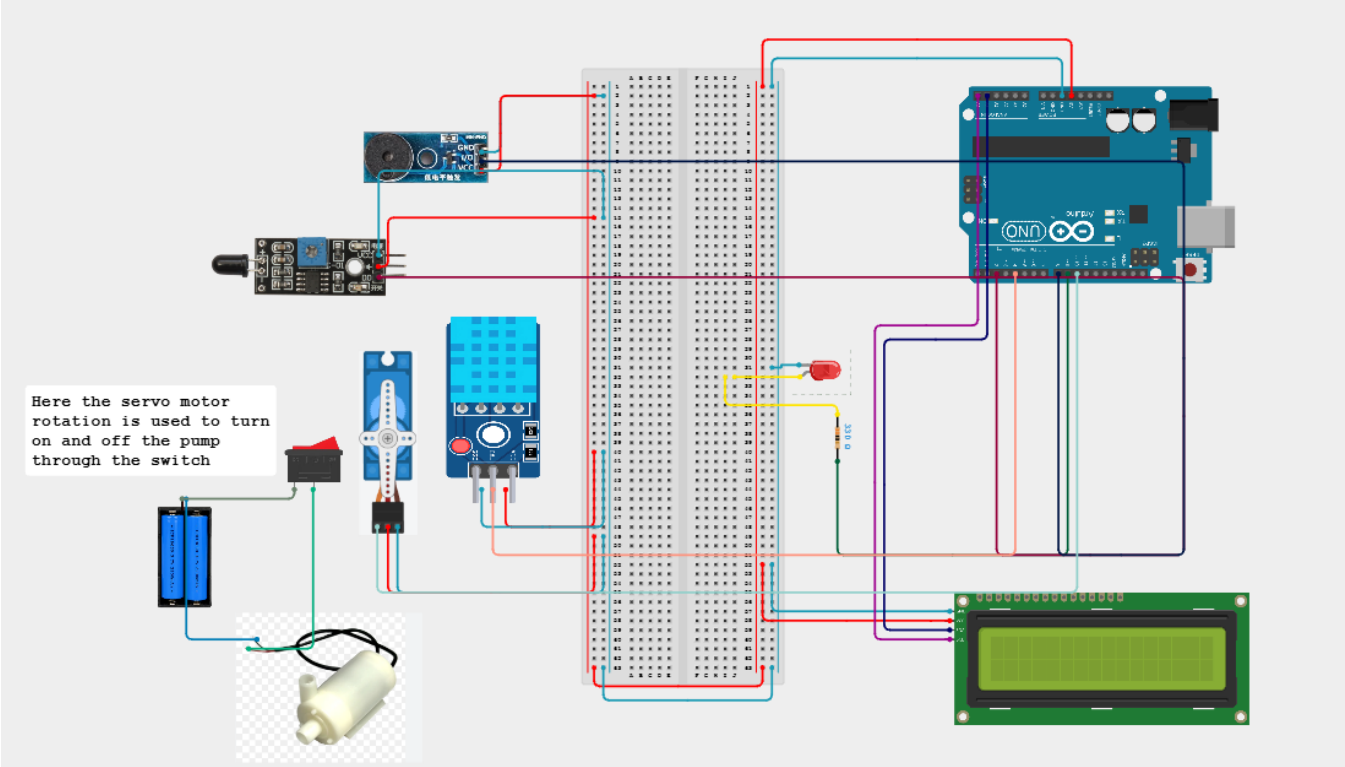


Figure: Functional connection diagram

Table: Connection table

DEVICE	Device Pins	Arduino connection
Flame sensor module	VCC / GND / DO	5 V / GND / D2
DHT11 module	VCC / GND / DATA	5 V / GND / D4
16×2 I2C LCD	VCC / GND / SDA / SCL	5 V / GND / A4 / A5
LED	Signal / return	D9 through resistor / GND
Buzzer module	VCC / GND / SIG	5 V / GND / D8
Servo motor	Signal / power / GND	D10 / 5 V / GND
Pump circuit (battery holder, rocker switch, pump)	Separate circuit	No electrical connection to Arduino pins

8.2 Representative Program Structure

#include <Servo.h>

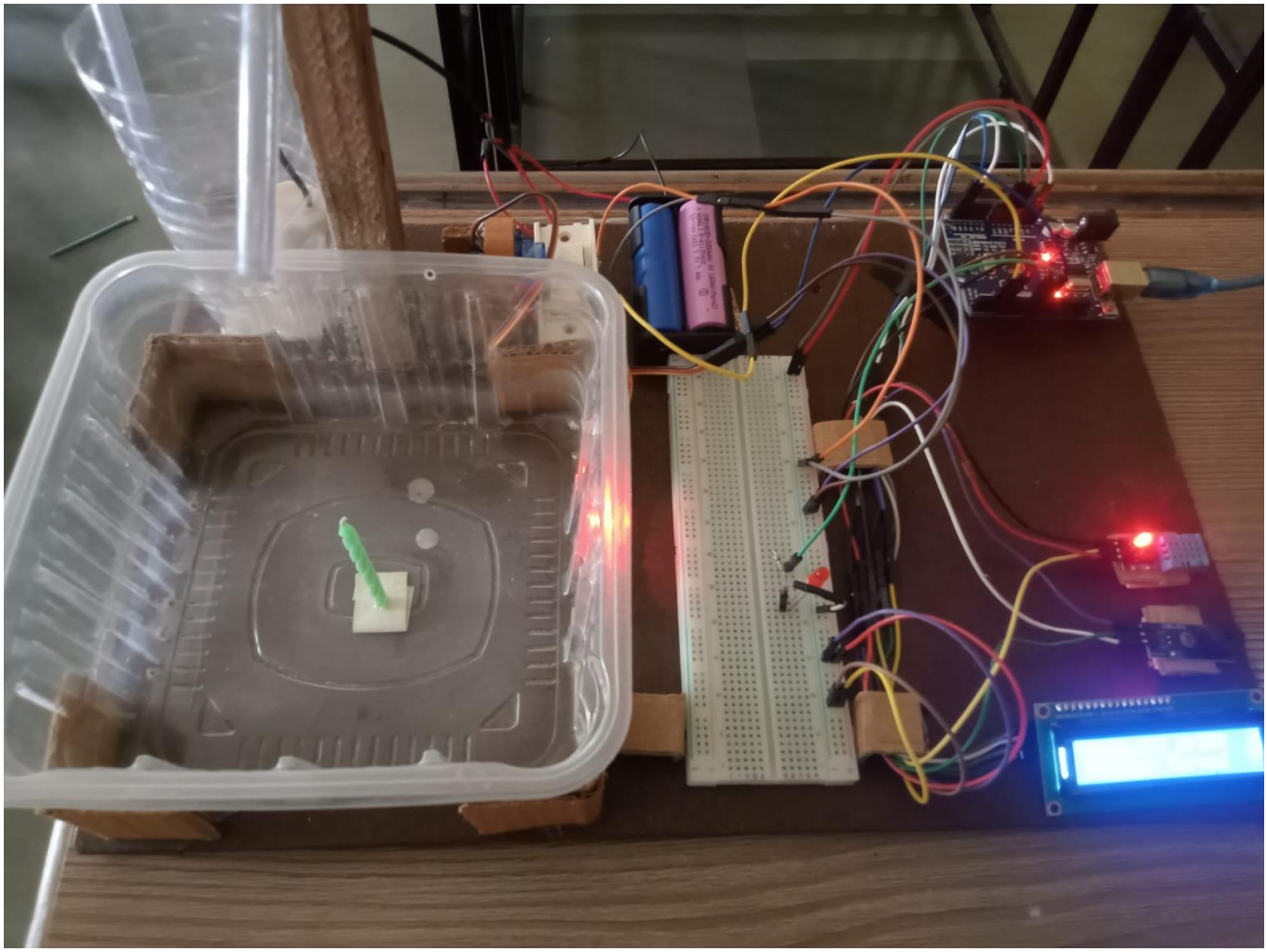
```
#include <DHT.h>
#include <LiquidCrystal_I2C.h>

const int FLAME_PIN  = D2;
const int LED_PIN    = D9;
const int BUZZER_PIN = D8;
const int SERVO_PIN  = D10;
const int DHT_PIN    = D4;
const int SERVO_OFF  = 120°;
const int SERVO_ON   = 60°;

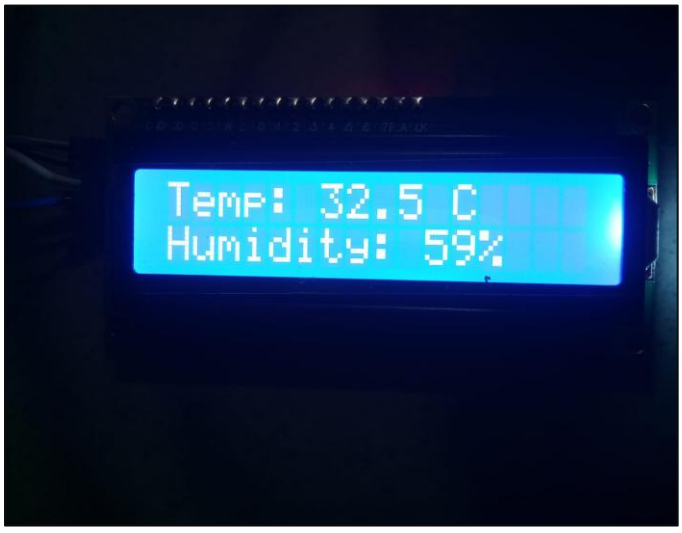
void setup () {
  // start LCD, DHT sensor, servo; set pin modes
  // move servo to SERVO_OFF
}

void loop () {
  bool flame = (digitalRead(FLAME_PIN) == LOW); // active LOW
  // read temperature and humidity from the DHT11
  if (flame) {
    // LED on, buzzer on, show fire alert, servo -> SERVO_ON
  } else {
    // LED off, buzzer off, servo -> SERVO_OFF
  }
  // update LCD with status, temperature, humidity
}
```

8.3 Photographs of the Prototype







CHAPTER 9: CONCLUSION

Problem. Students need a simple, low-cost physical model that combines fire sensing, local warnings, environmental monitoring and a basic automatic response. Existing research on fire detection is extensive, but many designs rely on communication modules or several sensors.

Proposed solution. This project builds a cardboard smart-house prototype around an Arduino Uno R3. It uses a flame sensor (active LOW), a DHT11 sensor, a 16×2 I2C LCD, an LED, a buzzer module, a servo motor, and a small submersible pump on a separate battery supply. The program is written in C++ in the Arduino IDE with the Servo, DHT and LiquidCrystal_I2C libraries.

Working principle. When the flame signal is detected, the LED and buzzer are switched on, a fire alert is shown on the LCD, and the servo moves the rocker switch to ON (60°), which connects the separately powered pump so that water flows through a tube. When the signal clears, the servo returns the switch to OFF (120°). Temperature and humidity are displayed for monitoring only.

Contribution. Fire detection already exists, so the project is not a new invention. Its contribution is an educational integration of available components, a servo-operated switch that keeps the pump circuit separate from the Arduino, and a record of practical problems met during development (relay behaviour, servo direction, and display corruption observed when the pump ran near the electronics, with the root cause not confirmed).

Limitations. The system depends on a single flame-sensor signal, has no smoke or gas sensing, has no network alerts or logging, uses temperature for display only, and is not certified for real-world fire protection. The water demonstration does not prove the ability to extinguish real fires. Improvements such as extra sensors, notifications, logging, and a more reliable suppression mechanism are future work (Section 7.5).

REFERENCES

The references are in the IEEE style. Items [3]– [12] are the research papers reviewed in Chapter 2; the others are a standard, datasheets and documentation, and the SDG source. Add the date of access to online items if the department requires it.

[1] Bureau of Indian Standards, "Selection, installation and maintenance of automatic fire detection and alarm system – Code of practice (Fourth Revision)," IS 2189:2008, New Delhi, India, Apr. 2008.

[2] Arduino, "Arduino UNO R3 datasheet (A000066)." [Online]. Available: https://docs.arduino.cc/resources/datasheets/A000066-datasheet.pdf

[3] A. Mahgoub, N. Tarrad, R. Elsherif, A. Al-Ali, and L. Ismail, "IoT-based fire alarm system," in Proc. 3rd World Conf. Smart Trends Syst., Secur. Sustain. (WorldS4), London, U.K., 2019, pp. 162–166, doi: 10.1109/WorldS4.2019.8904001.

[4] J. Wang, J. Su, and R. Hua, "Design of a smart independent smoke sense system based on NB-IoT technology," in Proc. Int. Conf. Intell. Transp., Big Data Smart City (ICITBS), 2019, doi: 10.1109/ICITBS.2019.00104.

[5] M. S. Dauda and U. S. Toro, "Arduino based fire detection and control system," Int. J. Eng. Appl. Sci. Technol. (IJEAST), pp. 447–453, Mar. 2020, ISSN 2455-2143.

[6]H. Alqourabah, A. Muneer, and S. M. Fati, "A smart fire detection system using IoT technology with automatic water sprinkler," Int. J. Elect. Comput. Eng., vol. 11, no. 4, pp. 2994–3002, 2021, doi: 10.11591/ijece. v11i4.pp2994-3002.

[7] M. C. Baba, J. B. Grado, D. J. L. Solis, I. M. Roma, and J. T. Dellosa, "A multisensory Arduino-based fire detection and alarm system using GSM communications and RF module with an Android application for fire monitoring," Int. J. Innov. Sci. Res. Technol., vol. 7, no. 3, pp. 964–968, Apr. 2022.

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[9] M. A. Al Rakib et al., "Fire detection and water discharge activity for fire fighting robots using IoT," Eur. J. Eng. Technol. Res., vol. 7, no. 2, pp. 128–133, 2022, doi: 10.24018/ejeng.2022.7.2.2742.